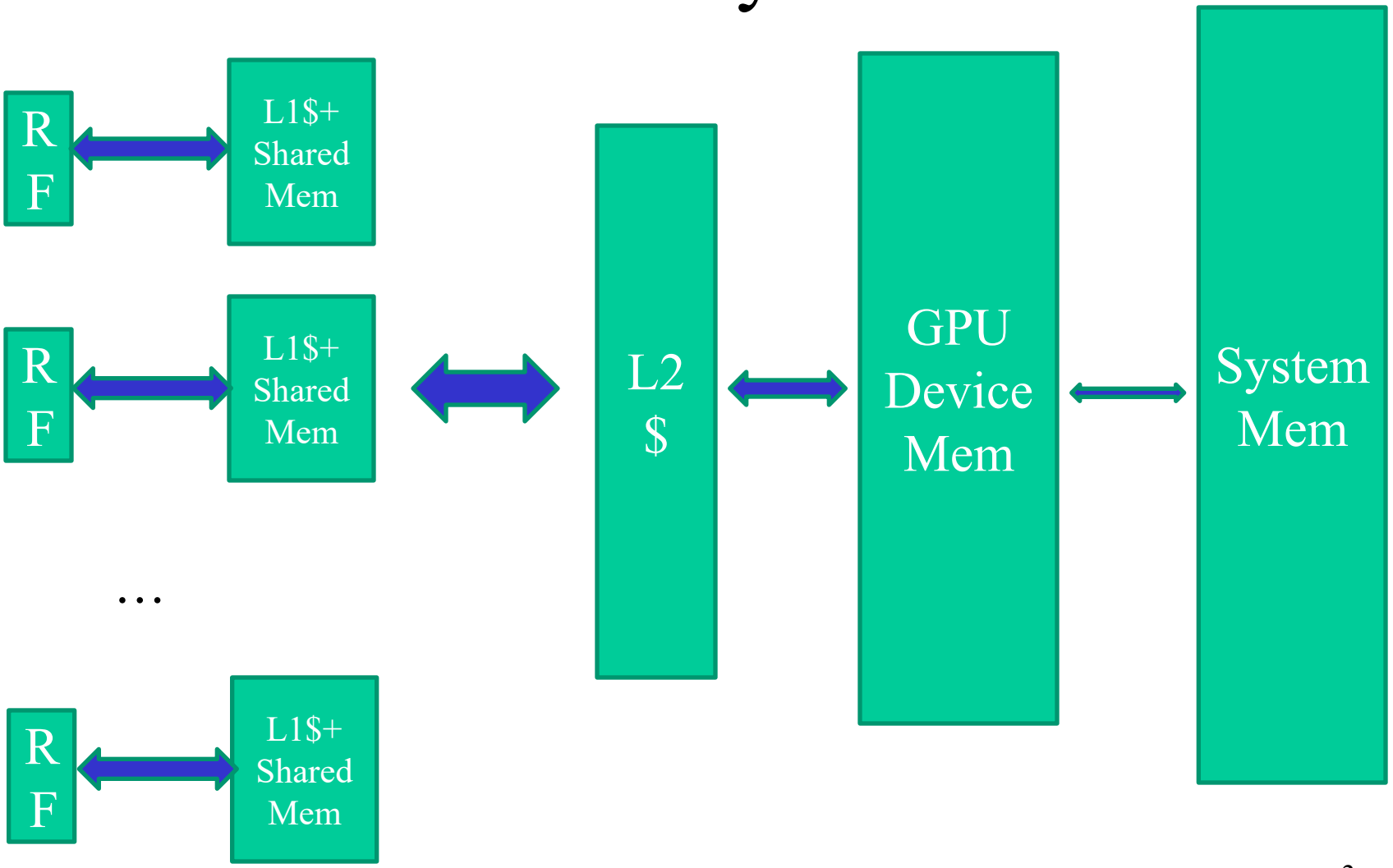
GPU Memory Architecture

Outline

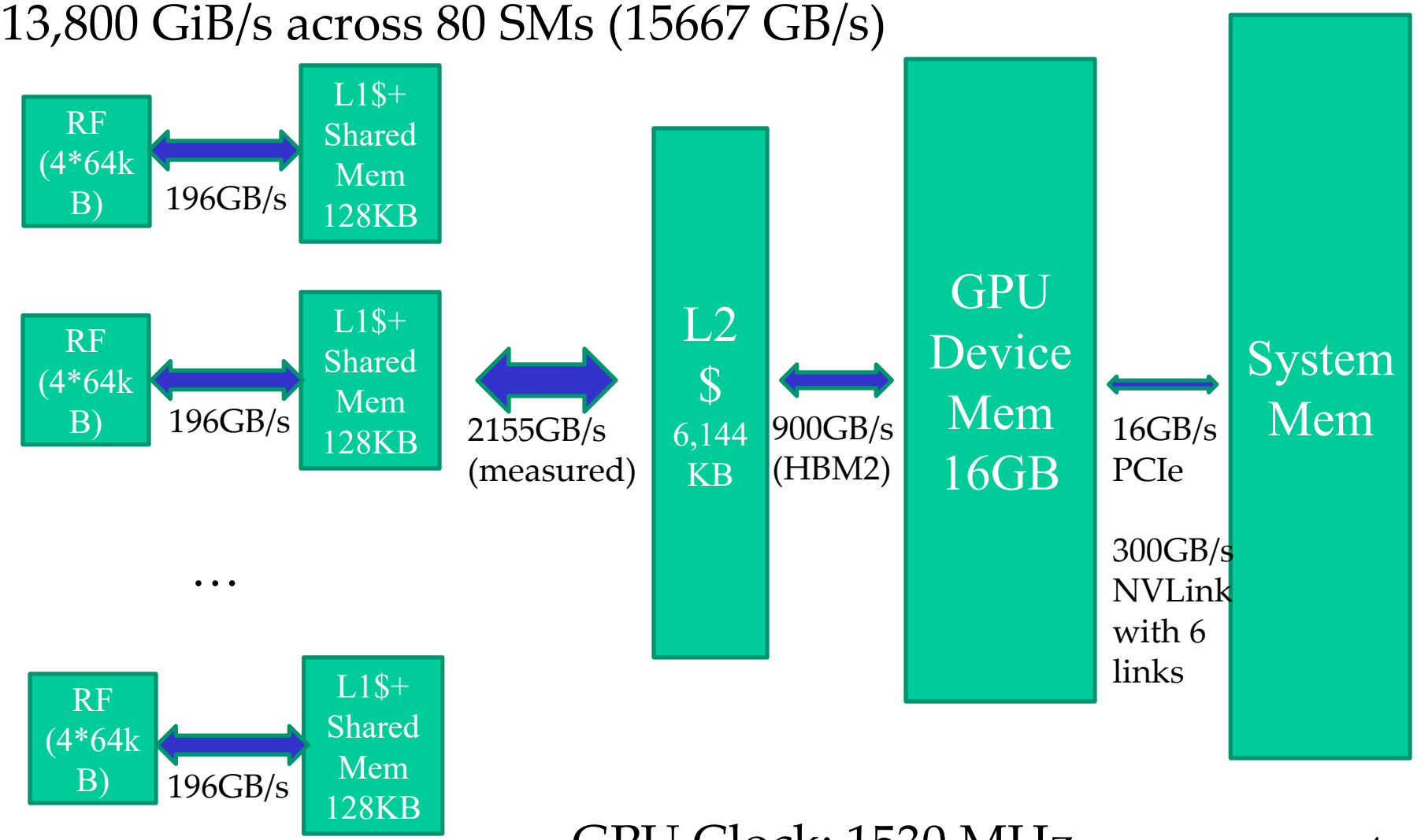
- GPU Memory Architecture
- Memory Bandwidth Optimization
  - Coalescing
- L1 D-cache vs. Shared Memory
  - Shared memory bank conflicts
- GPU Memory Model (Software side)
  - Global memory
  - Local memory
  - Shared memory usage
    - Caching frequently used data
    - Enabling memory coalescing
- GPU Paging Support

Memory Hierarchy in Throughput-Oriented Systems



GV100 (Volta) Peak BW

13,800 GiB/s across 80 SMs (15667 GB/s)



GPU Clock: 1530 MHz

Volta Memory Hierarchy

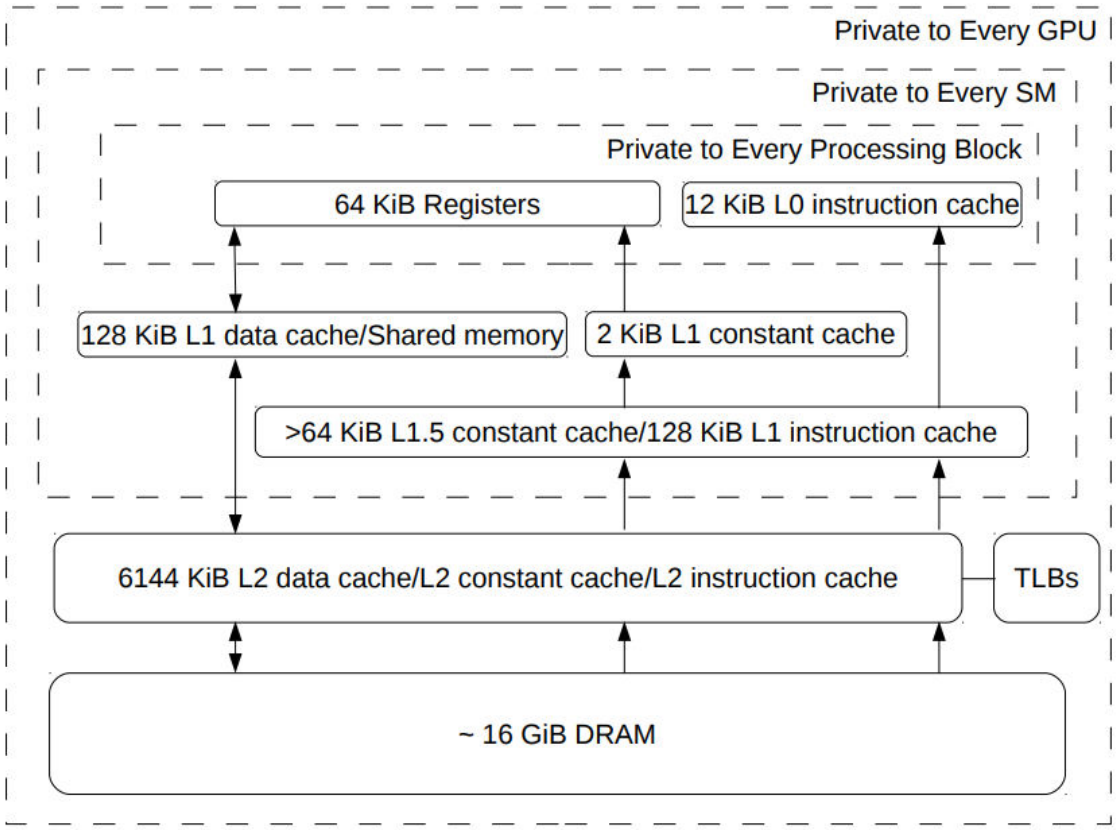


Figure 3.1: Memory hierarchy of the Volta V100 GPU (GV100).

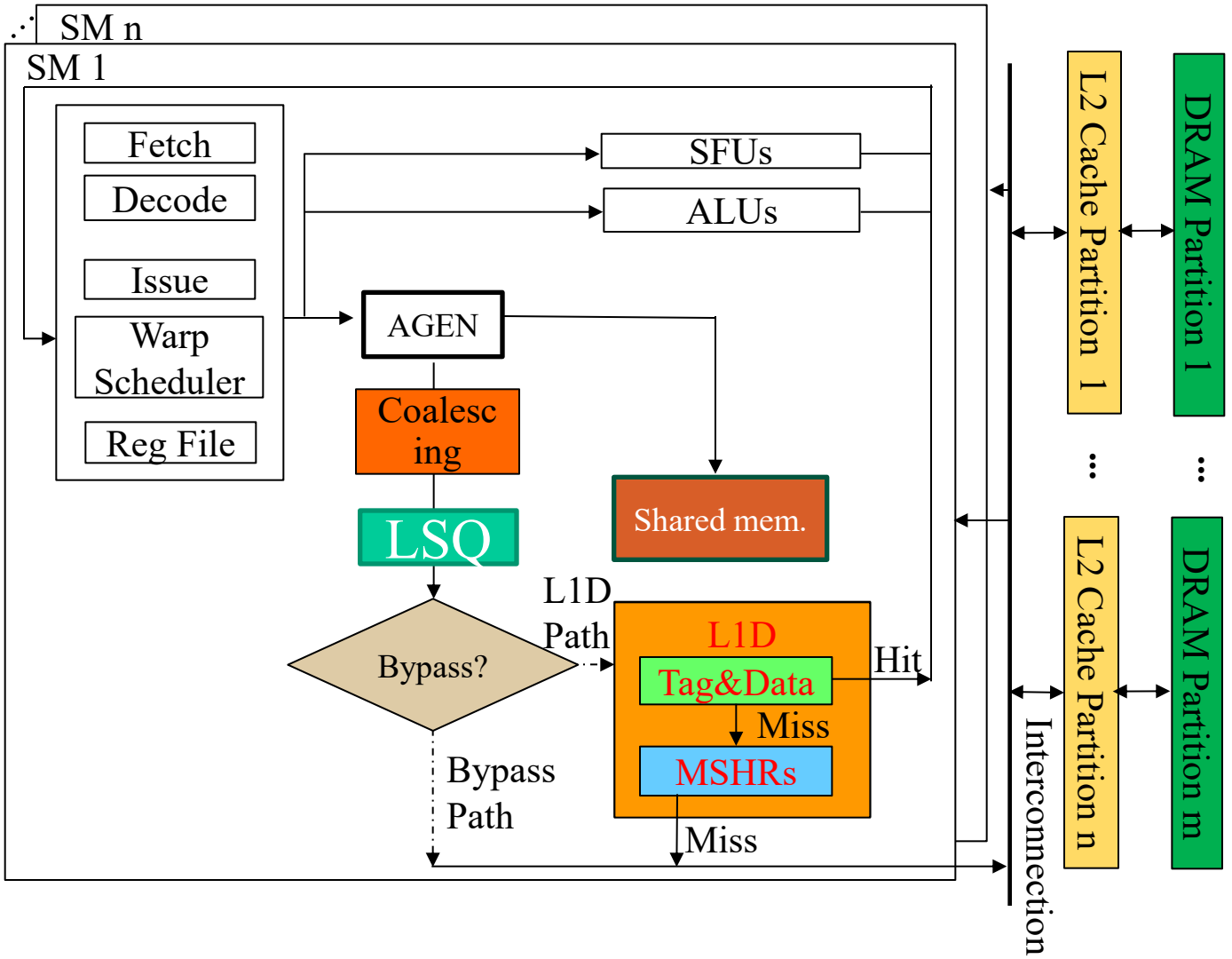
"Dissecting the NVIDIA Volta GPU Architecture via Microbenchmarking" by Jia et al, 2018

Volta SM Architecture



Figure 5. Volta GV100 Streaming Multiprocessor (SM)

Memory Hierarchy in an SM

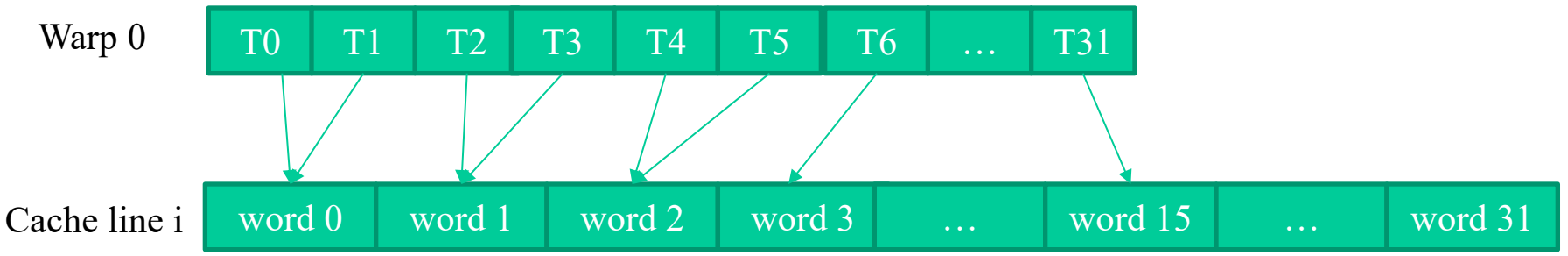


Coalescing

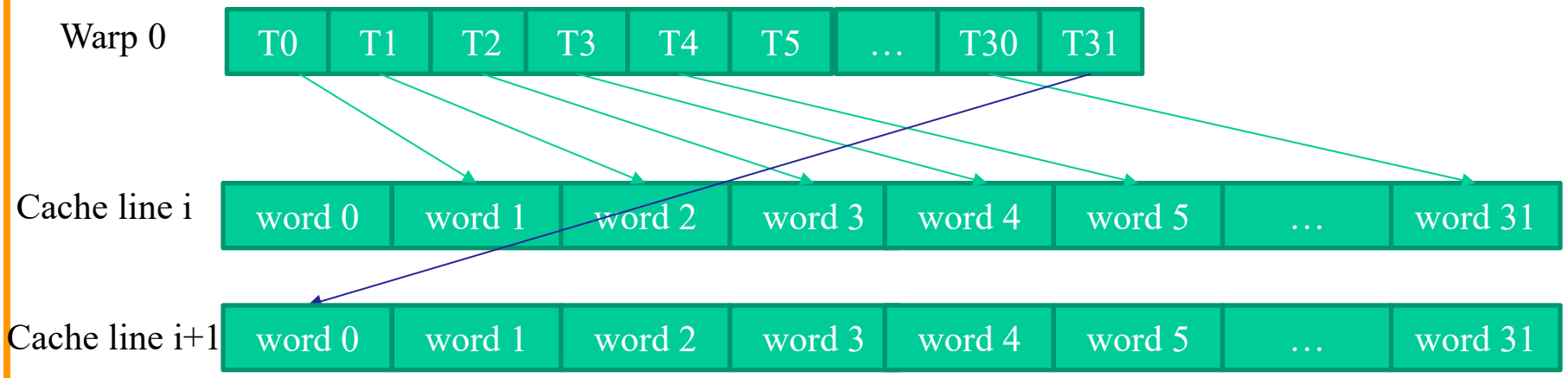
- Combining memory accesses made by threads in a warp into transactions at the cache line granularity (128B)
  - E.g. if each thread in a warp accesses a 4-byte consecutive data
    - Send one 128–byte request to L1 D-cache/Interconnect instead of 32 requests.
- This reduces the number of transactions between SIMT cores and memory system
  - Less work for Interconnect, Memory Partition and DRAM.

Examples

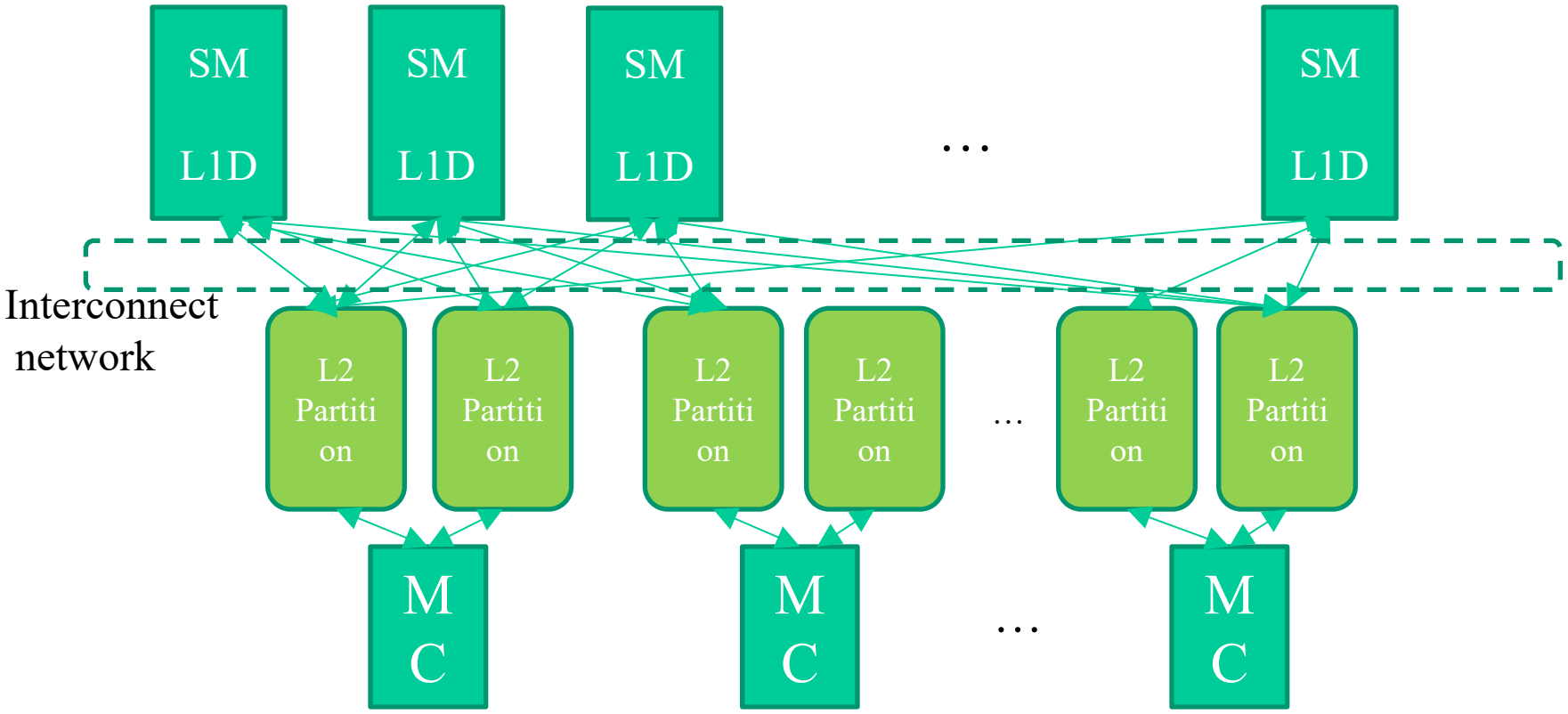
A[tid/2]: coalesced access assuming A is aligned to 128-byte boundary



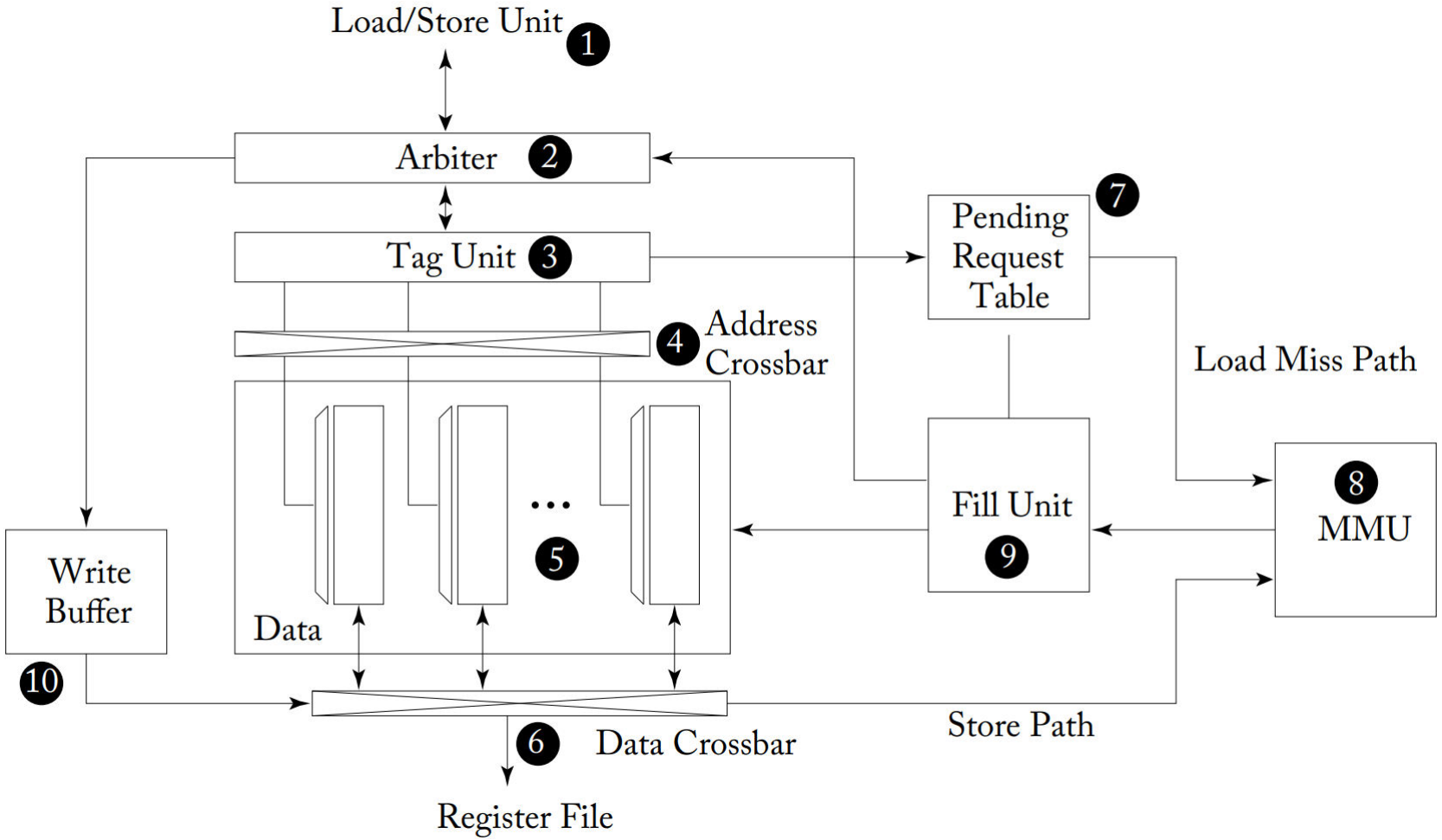
A[tid+1]: uncoalesced access assuming A is aligned to 128-byte boundary



L1, L2, and Beyond



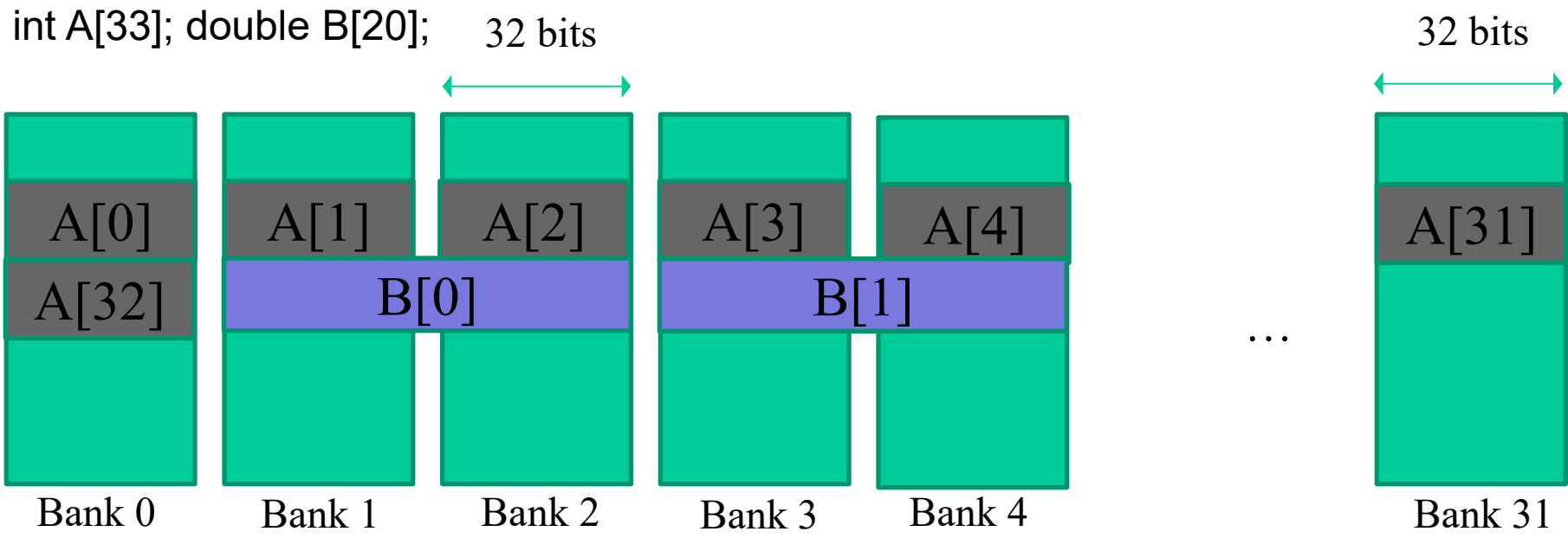
Unified L1 data cache and shared memory

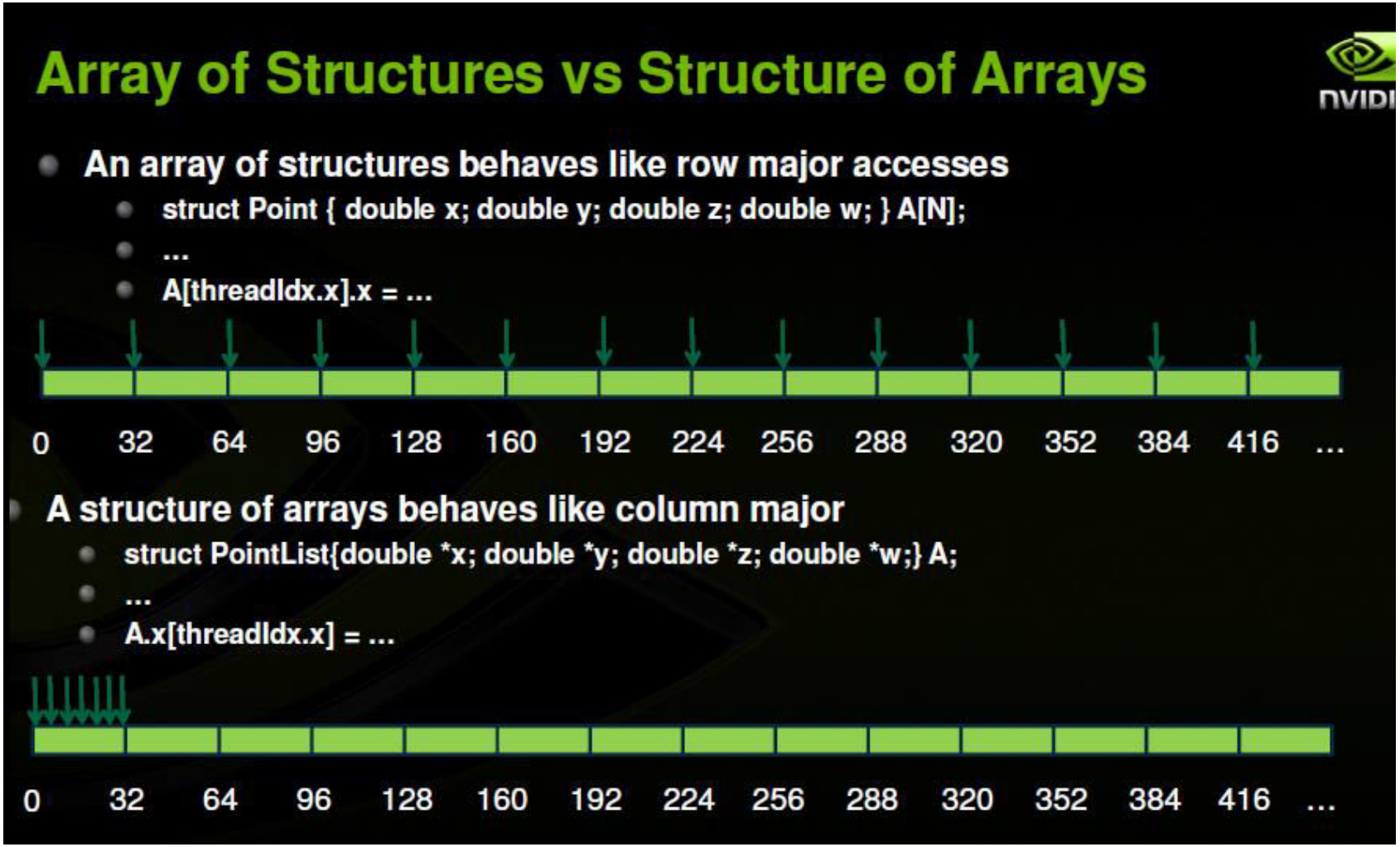


[Minkin et al., 2012].

Shared Memory

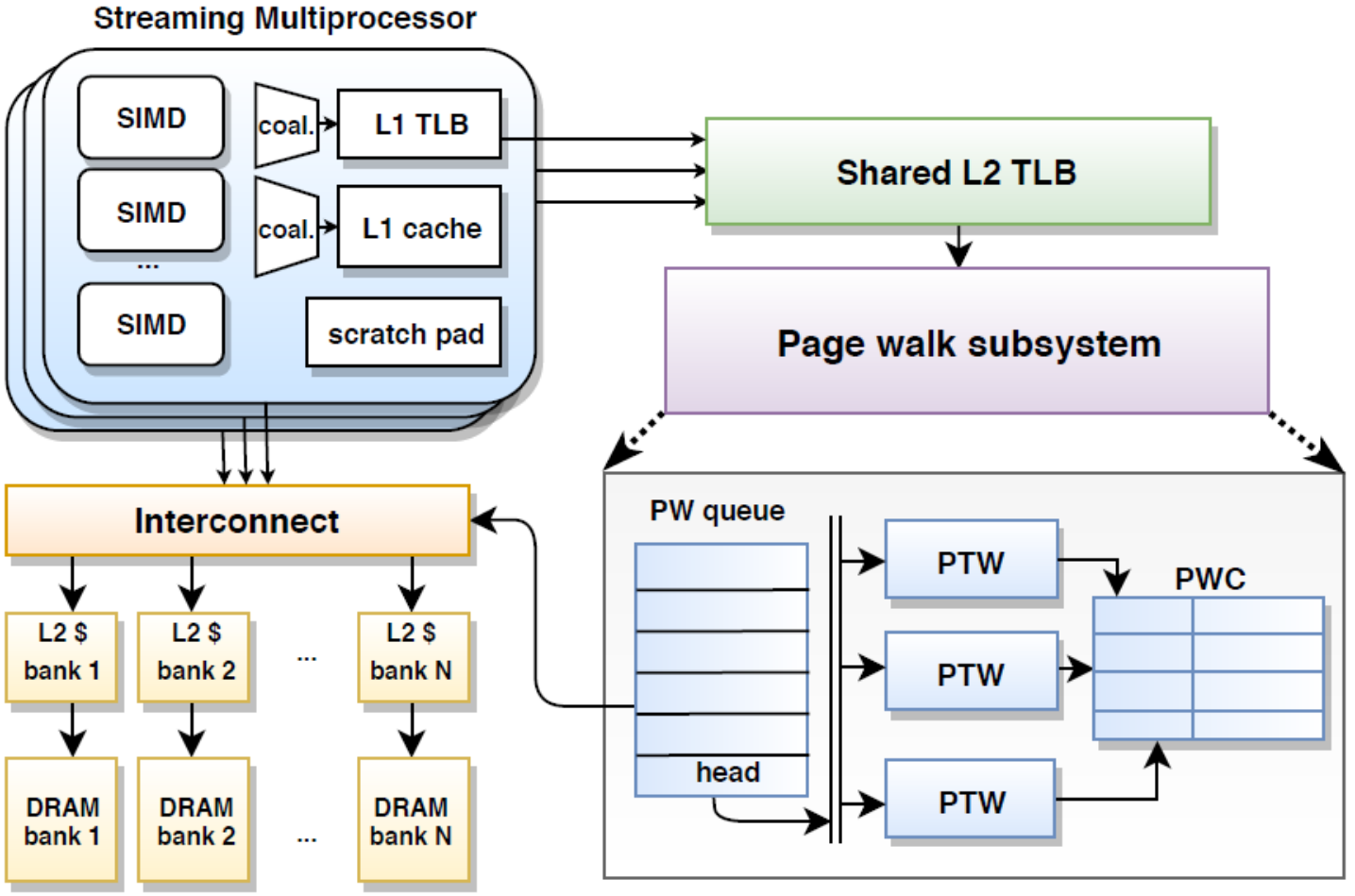
- Multi-banked structure for high BW.
  - One read port and one write port per bank.
- Explicit software management.
- Data layout in shared memory
  - Identify and handle bank conflicts



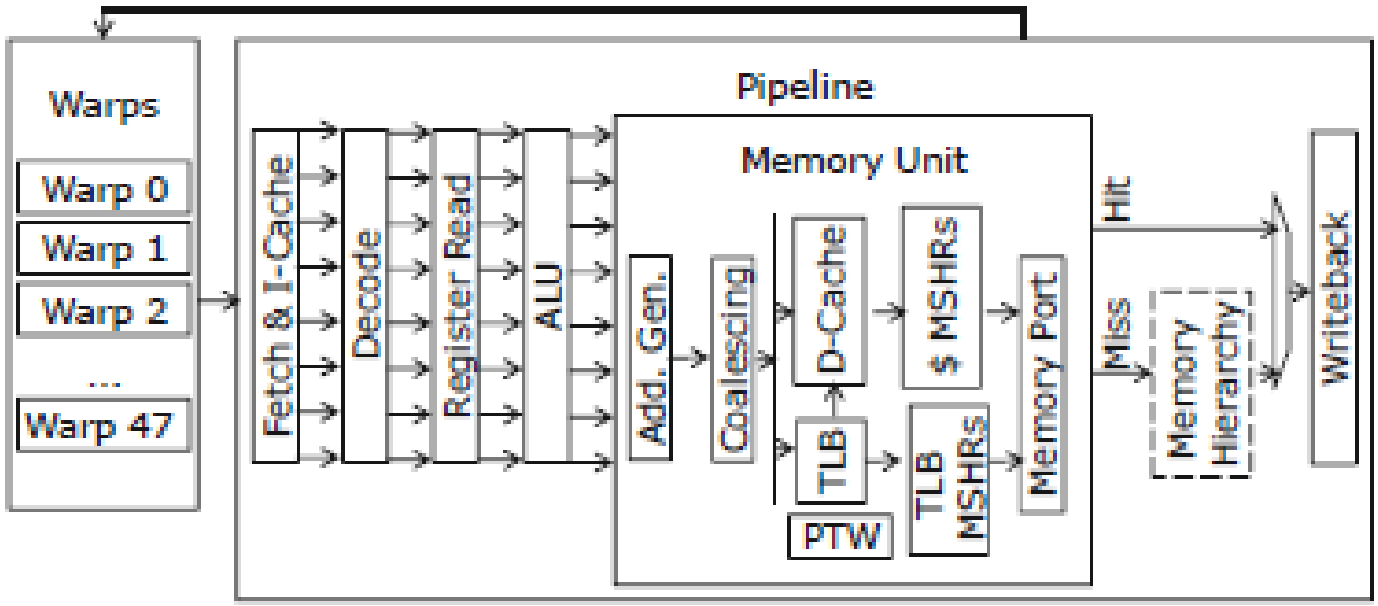


Considering AoS vs. SoA for data in global memory and data in shared memory

GPU MMU

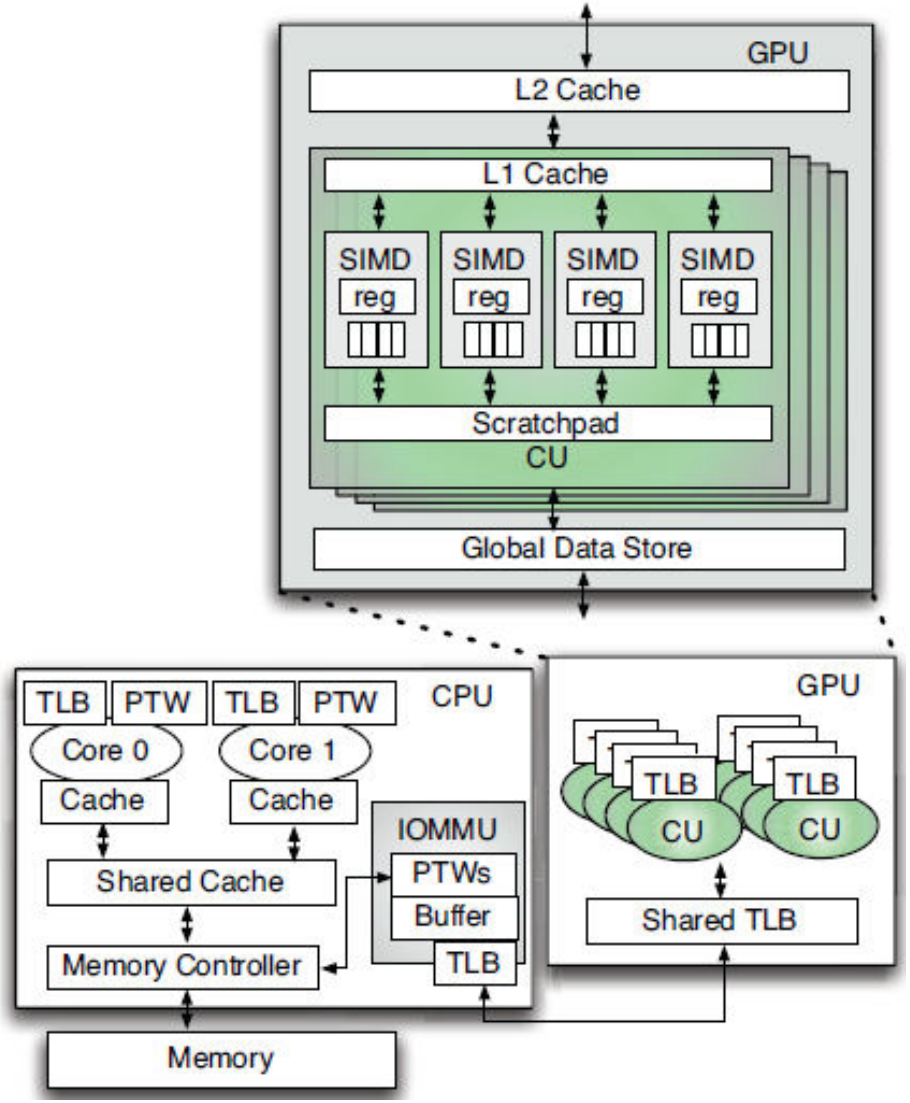

Streaming Multiprocessor
SIMD
SIMD
...
SIMD
coal.
coal.
L1 TLB
L1 cache
scratch pad
Shared L2 TLB
Page walk subsystem
Interconnect
L2 $ bank 1
L2 $ bank 2
...
L2 $ bank N
DRAM bank 1
DRAM bank 2
...
DRAM bank N
PW queue
head
PTW
PTW
PTW
PWC

GPU Address Translation



"Architectural support for address translations on GPUs", ASPLOS14

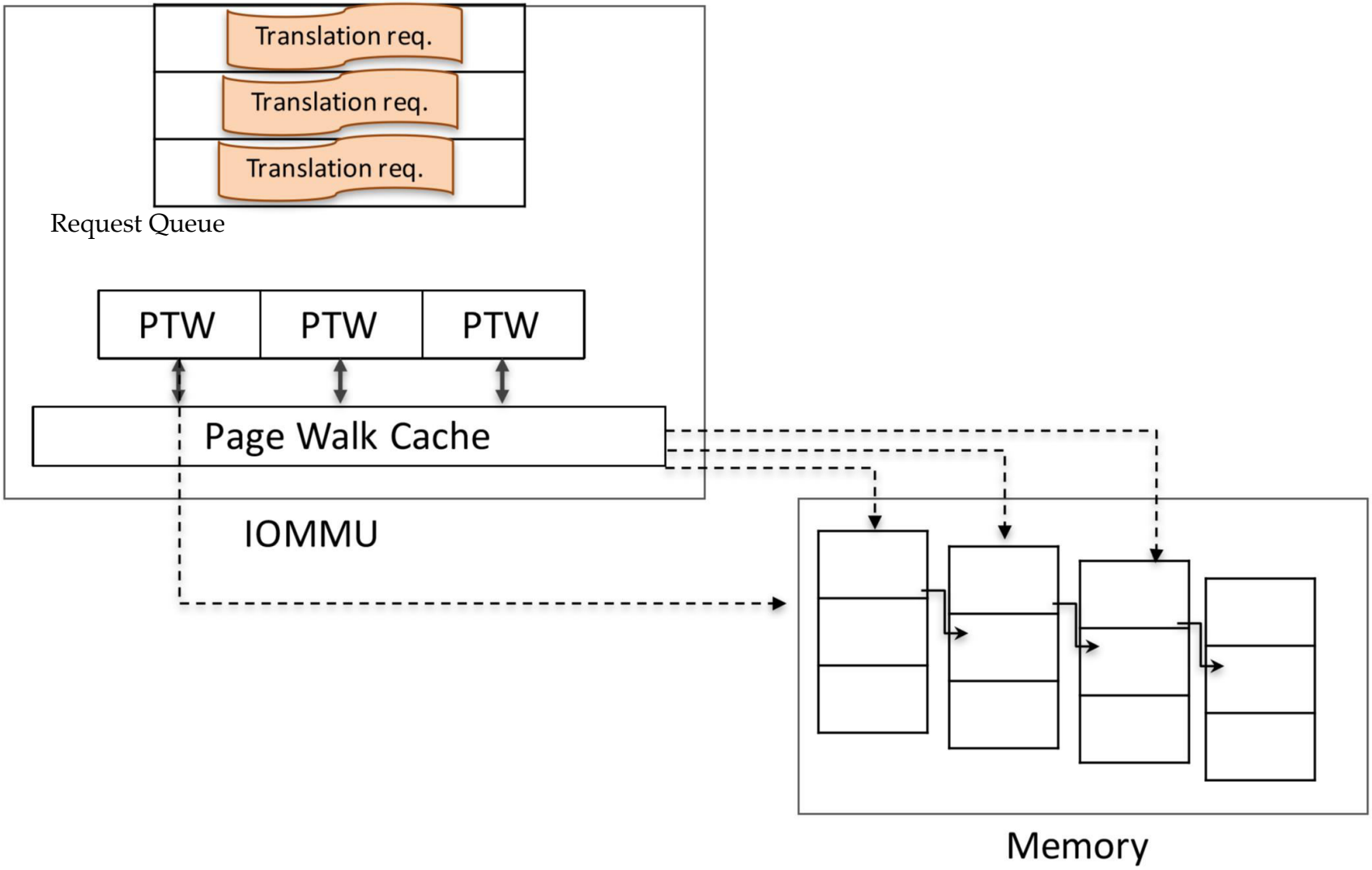
Supporting Unified Memory



Life of a GPU Address Translation Request:
1 An address translation request is generated when executing a SIMD memory instruction (load/store).
2 A coalescer merges multiple requests to the same page (e.g., 4KB) generated by the same SIMD instruction.
3 The coalesced translation request looks up the GPU's L1 TLB and then the GPU's shared L2 (if L1 misses).
4 On a miss in the GPU's L2 TLB, the request is sent to the IOMMU on the CPU.
5 Upon arrival at the IOMMU, the request looks up the IOMMU's TLBs.
6 On a miss, the request queues up as a page walk request in the IOMMU buffer.
7 When an IOMMU's page table walker becomes free, it typically selects a pending request from the IOMMU buffer in FCFS order.
8 The page table walker first performs a PWC lookup and then completes the walk of the page table, generating one to four memory accesses.
9 On finishing a walk, the desired translation is returned to the TLBs and ultimately to the SIMD unit that requested it.

"Scheduling Page Table Walks for Irregular GPU Applications", ISCA18

CPU IOMMU


Translation req.
Translation req.
Translation req.
Request Queue
PTW
PTW
PTW
Page Walk Cache
IOMMU
Memory

Supporting Unified Memory

1. CU accesses memory location causing TLB miss
2. TLB miss is relayed to GMMU
3. Page is not present (far-fault). GMMU allocates MSHR entry
4. GMMU sends 'NACK-replayable' message to requesting TLB
5. CPU is interrupted to handle page migration & page table update
6. Page migration complete. GMMU notifies TLB to replay request

